PREDICT-THEN-OPTIMIZE FRAMEWORK FOR PUBLIC TRANSPORT LINE REDESIGN UNDER FLUCTUATING TRAFFIC CONDITIONS

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Abstract: Public Transport (PT) lines are traditionally designed to optimize performance under nominal traffic conditions. This paper proposes a method to proactively redesign appropriate parts of PT networks when faced with high traffic fluctuations. We adopt a predict-then-optimize paradigm in which PT lines are reconfigured based on traffic forecasts via an NSGA-III-based algorithm. On the Beijing network under high variability, average travel time improves by up to 25.8% while preserving over 85% line overlap; unlike stop-skipping baselines, which break connectivity for a large share of OD pairs, our redesign preserves full OD connectivity.

Keywords: Public Transport network redesign; Predict-then-optimize; Multi-objective optimization; Travel time uncertainty; Structural continuity

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1. Introduction

Public Transport (PT) networks are typically planned for nominal, scheduled conditions (Ibarra-Rojas et al., 2015), while real-world operations are continuously subject to variability caused by congestion and other factors (Büchel & Corman, 2020), especially in mixed traffic environments.

Existing approaches to cope with recurrent perturbations can be broadly classified as *preventive*, *reactive*, or *proactive*, depending on the decision horizon and intervention level. Reactive strategies attempt, at the operational level, to restore service after perturbations have occurred, for instance via stop-skipping, holding, or detouring (Peled et al., 2021).

A key limitation shared by both preventive and reactive strategies is that the PT network structure typically remains fixed, despite persistent changes in operating conditions. Proactive approaches aim to overcome this limitation by anticipating variability through short-term predictions and revising the line design before the next operating interval (Peled et al., 2019).

Most existing studies treat PT network design as a deterministic optimization problem, ignoring the inherent uncertainty and variability in urban traffic conditions. An exception is Guo et al. (2025), where, however, (i) future travel times on road links were assumed to be perfectly known (which is unrealistic), so the impact of prediction errors was not considered, while we model the error of a well established prediction model, (ii) only a single simplified objective was adopted, while we solve a multiobjective optimization, and (iii) their approach is tested only on a simple hypothetical scenario, while we consider real-world dataset to fit the prediction error models.

The **main contribution** of this paper is to assess the potential of predict-then-optimize proactive redesign of bus networks. To this aim, we propose a new method that adapts bus lines to time-varying stochastic traffic congestion. We solve a multi-objective problem to minimize operational cost and user travel time, and to limit structural network modifications, so as to ease operations and user routing. Adaptation is triggered only when forecasted conditions severely deviate from the nominal ones; moreover, such adaptation is anticipated (e.g., 1h in advance) to avoid disrupting journeys of current travelers.

In the Beijing large scale network, beyond a one-off static re-optimization gain of 11.4%, with no link travel time variability, the benefits grow with such a variability. When it is very high, average travel time improves by up to 25.8% and operational cost by up to 22.6%, while line overlap stays above 85% (Table 2). Compared with a stop-skipping baseline, which improves travel time by at most 19.7% but only by disconnecting 28–39% of OD pairs under high variability, our redesign attains a larger gain while keeping every OD pair reachable. An extended version of this paper, with the full literature review, proofs, algorithmic details and additional experiments, is available as an extended report (Guo et al., 2026).

2. Model and Problem Formulation

As illustrated in Fig. 1, our methodology follows a predict-then-optimize framework that explicitly integrates current observations into prediction. To ensure timely responsiveness, this procedure is repeated periodically, i.e., once per hour. At the start of each time slot, the new network is announced in advance (e.g., 1h), so passengers can plan a route that remains valid for their entire trip and is never changed mid-journey; we assume here that the longest trip is shorter than one time slot (1h).

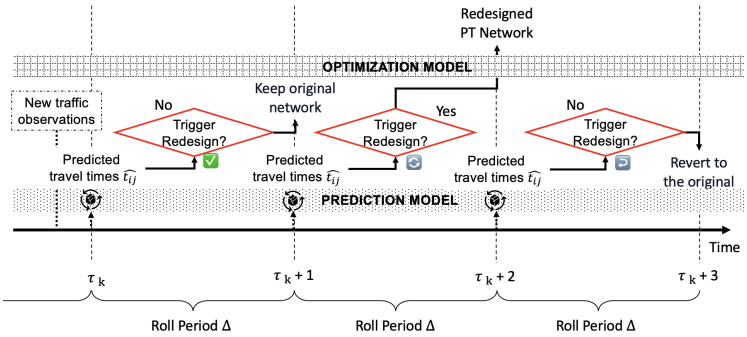


Figure 1: Proactive Line Redesign Framework.

2.1 Graph Model of PT Network

We represent the underlying road infrastructure as an undirected substrate network $\mathcal{G}^{\text{sub}} = (\mathcal{V}^{\text{sub}}, \mathcal{E}^{\text{sub}})$, where $\mathcal{V}^{\text{sub}}$ are candidate bus stops and $\mathcal{E}^{\text{sub}} = \mathcal{V}^{\text{sub}} \times \mathcal{V}^{\text{sub}}$ the potential connections among them. Let $\mathcal{E}$ denote the set of feasible edges in the network. Let $\mathcal{L}$ denote the set of all PT lines in the network. For each PT line $\ell \in \mathcal{L}$, headway h^ℓ can be calculated by $h^\ell = \frac{T^\ell}{n^\ell}$, where T^ℓ represents the round trip time along the entire line, and n^ℓ denotes the number of vehicles operating on that line (Eq. 4.4 of Vuchic et al., 1981). Assuming random passenger arrivals, the average waiting time for line ℓ is given by

$$w^\ell = \frac{h^\ell}{2}, \text{ where } h^\ell \text{denotes the service headway of line } \ell. \tag{1}$$

2.2 Stochastic Modeling of Travel Times

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For the sake of realism, we model the aforementioned bus travel times t_{ij} from stop i to stop j, for each edge $(i,j) \in \mathcal{E}^{\text{sub}}$, at a given time of day, as random variables. Let $t_{ij}^\omega, \omega \in \Omega$ denote a realization of this variable. We assume that a prediction model provides predictions $\hat{t}_{ij}$ for each edge. Each OD pair $k \in \mathcal{K}$ is defined by an origin stop $o_k \in \mathcal{V}^{\text{sub}}$ and a destination stop $d_k \in \mathcal{V}^{\text{sub}}$.

For any network configuration $\mathcal{G}_{\text{PT}}$ and any realization $\omega \in \Omega$, each link (i, j) can be traversed in time t_{ij}^{ω}. Based on such values, we compute shortest paths for any OD pair $k \in \mathcal{K}$ and denote the trip travel time as t_k^{ω}. Note that t_k^{ω} includes (i) in-vehicle travel times, (ii) dwell times ($d = 1$ min per stop), (iii) initial waiting times, and (iv) transfer waiting times plus the fixed 5 min transfer penalties. When computing shortest paths we assume links are uncongested: link travel times are exogenous to passenger volumes, so the reported trip times are user-optimal times that abstract away vehicle capacity and in-vehicle crowding (a redesigned line is never penalized for the demand it attracts).

In our framework, three notions of OD travel time are defined: (i) the actual travel time t_k, described in the paragraph above; we consider its realizations when evaluating the performance of our method. (ii) the predicted travel time $\hat{t}_k$, computed as the sum of the predicted link times $\hat{t}_{ij}$ along the shortest paths described in the paragraph above; we use these predicted values as input parameters of optimization (4); (iii) the baseline travel time t_k^0, defined as the OD travel time on the original network design $\mathcal{G}_{\text{PT}}^{\text{org}}$ under baseline link times $t_{ij}^0, \forall (i, j \in \mathcal{E})$.

2.3 Problem Statement

This research tackles the proactive re-optimization of public transport lines (e.g., one hour ahead), triggered when traffic conditions heavily deviate from the nominal ones, in order to preserve user and operator performance.

2.3.1 Timeframe and Stability Considerations

The operational feasibility of the underlying assumptions—one-hour advance notice, predefined deviation zones, and app-based passenger guidance—is context-dependent. They will be more realistic in settings where flexible routing will be organizationally and technologically supported (e.g., with forthcoming vehicle automation (Peled et al., 2019)), while in more rigid operating environments the results presented here should be interpreted as upper bounds on the performance gains that could be achieved as operational practices, vehicle automation, and passenger-information systems evolve.

Although PT line reconfiguration is implementable, PT lines are more easily manageable if disruptive reconfigurations are limited. For this reason, we measure structural stability using the Jaccard index (Jaccard, 1901), defined for the edge sets of the original and redesigned networks as

$$J(\mathcal{E}_{\text{orig}}, \mathcal{E}_{\text{new}}) = \frac{|\mathcal{E}_{\text{orig}} \cap \mathcal{E}_{\text{new}}|}{|\mathcal{E}_{\text{orig}} \cup \mathcal{E}_{\text{new}}|} \in [0, 1]. \tag{2}$$

As shown in Fig. 1, we use the following hourly trigger.

Rule 1 (Forecast-triggered line redesign)**.** *At the start of a time slot, define* $\bar{\Delta} \stackrel{\text{def}}{=} \frac{1}{|\mathcal{E}^{sub}|} \sum_{(i,j)\in\mathcal{E}^{sub}} \frac{|\hat{t}_{ij} - t_{ij}^0|}{t_{ij}^0}$. *For a monitoring tolerance* $\delta_{\text{trig}} \in [0, 1]$*, the announced network is kept if* $\bar{\Delta} \leq \delta_{\text{trig}}$*. If* $\bar{\Delta} > \delta_{\text{trig}}$*, the operator solves* (4) *using the predicted link times* $\hat{t}_{ij}$*, and the selected solution becomes* $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ *for the next one-hour slot, subject to operator approval.*

The rule determines *whether* a redesign is attempted. Conditional on triggering, the extent of structural change is controlled by Z_3 in (4).

The Jaccard-based structural term also provides a simple stability guarantee.

Proposition 2 (Horizon-independent pairwise stability)**.** *Let* $\mathcal{E}^{(k)}$ *denote the set of active edges in the redesigned network at step k, and* $\mathcal{E}^{(0)}$ *for the original network. Then, for any two redesign steps k and* $k+n$ *(*$n \geq 1$*), it holds that*[1]

$$J\big(\mathcal{E}^{(k)}, \mathcal{E}^{(k+n)}\big) \;\geq\; J\big(\mathcal{E}^{(k)}, \mathcal{E}^{(0)}\big) + J\big(\mathcal{E}^{(k+n)}, \mathcal{E}^{(0)}\big) - 1. \tag{3}$$

[1]The proof is provided in the extended report (Guo et al., 2026).

2.3.2 Objective Functions

We formulate our public transport network adaptation problem as a three-objective optimization, as expressed in (4). The first objective, Z_1, minimizes passenger travel time deterioration, where q_k represents the travel demand flow. The second objective Z_2 minimizes operational costs through total vehicle-hours, where $\hat{t}^\ell$ is the predicted one-way travel time of line ℓ (in-vehicle plus dwell times) under link times $\hat{t}_{ij}$.

$$\min_{\mathcal{G}_{\mathrm{PT}}^{\mathrm{adpt}}} \begin{pmatrix} Z_1 \\ Z_2 \\ Z_3 \end{pmatrix} = \begin{pmatrix} \underbrace{\sum_{k\in\mathcal{K}} q_k \cdot \max(\hat{t}_k - t_k^0, 0)}_{\text{Passenger travel time deterioration}} \\ \underbrace{\sum_{\ell\in\mathcal{L}} n^\ell \cdot 2\hat{t}^\ell}_{\text{Total vehicle-hours (operating cost)}} \\ \underbrace{1 - \dfrac{\sum_{\ell\in\mathcal{L}}\sum_{(i,j)\in\mathcal{E}} \min(e_{ij}^\ell, e_{ij}^{\ell,0})}{\sum_{\ell\in\mathcal{L}}\sum_{(i,j)\in\mathcal{E}} \max(e_{ij}^\ell, e_{ij}^{\ell,0})}}_{\text{Line dissimilarity}} \end{pmatrix} \tag{4}$$

In (4), $e_{ij}^{\ell,0}$ and e_{ij}^{ℓ} denote whether line ℓ uses substrate link (i,j) in the original and the redesigned network, respectively. The design is further subject to line-structure constraints (each active line is a simple path between two distinct terminals, enforced through flow-balance, degree and Miller–Tucker–Zemlin subtour-elimination constraints) and to a global fleet budget $\sum_{\ell\in\mathcal{L}} n^\ell \le V_{\max}$; the complete formulation is given in the extended report (Guo et al., 2026).

3. Solution Algorithm

Given the tri-objective nature of the problem and the nonconvex trade-offs, we adopt a Non-dominated Sorting Genetic Algorithm III (NSGA-III) framework (Deb & Jain, 2014) to approximate the Pareto front. We adopt a **design–evaluation decomposition** for computational tractability. The evolutionary search operates on network design variables (line topologies e_{ij}^ℓ and fleet allocations n^ℓ). The crossover exchanges complete lines between parents as the primary recombination mechanism; any subsequent modifications are limited to feasibility restoration. With probability P_m, a mutation perturbs a single line ℓ via local structural modifications, including node insertion, node deletion, or segment reconnection. At initialization, candidate networks are screened so that the origin and destination of every OD pair are connected; full feasibility is subsequently evaluated together with the objectives.

For each individual, the algorithm first constructs a comprehensive virtual-node graph $\mathcal{G}'$ that integrates in-vehicle travel times and waiting times w^ℓ, the latter being derived from the allocated vehicle units n^ℓ. To evaluate the travel-time deterioration Z_1, the algorithm solves a shortest-path problem (Dijkstra) for every OD pair $(o_k, d_k) \in \mathcal{K}$ on $\mathcal{G}'$. The resulting shortest path yields the predicted OD travel time $\hat{t}_k$. Specifically, Z_3 is quantified using a line-specific Jaccard similarity, where each network element is defined as a triplet $(\min(i,j), \max(i,j), \ell)$. Parameter values, the pseudo-code, the repair operators and a comparison with other multi-objective algorithms are reported in the extended report (Guo et al., 2026).

4. Evaluation Method

On each link (i,j), we generate travel time realization t_{ij}^ω by performing Monte-Carlo sampling from random variable $t_{ij} \sim \text{log-normal}(\mu_{ij}, \sigma_{ij}^2)$ such that the median of t_{ij} corresponds to the nominal time reported in the corresponding network dataset, and the coefficient of variation (cv) matches a predetermined value: we test cv = 0.05, 0.25, 0.5, 1.0.

We use the Diffusion Convolutional Recurrent Neural Network (DCRNN) (Li et al., 2017) as a representative spatio-temporal predictor to generate empirical one-hour-ahead prediction errors. Due to the

absence of historical traffic data for Mandl's Swiss network, the DCRNN is trained on the METR-LA dataset (Li et al., 2017), which contains traffic speed measurements from 207 sensors in Los Angeles County collected at 5-minute intervals over four months. For the real-world Beijing case study, instead, we utilize the Q-Traffic dataset (Liao et al., 2018), which provides traffic speed and road network data for the 6th Ring Road in Beijing, China.

To assess the sensitivity of our method to the magnitude of the prediction errors, in our optimization we set predicted travel times for a certain realization $\omega \in \Omega$ as

$$\hat{t}_{ij}^{\omega}(\alpha) = t_{ij}^{\omega} \exp\big(\alpha\, \eta_{ij}^{\omega}\big), \tag{5}$$

where η_{ij}^{ω} is sampled with replacement from the set of empirical logarithmic prediction errors $\eta_{ij} \stackrel{\text{def}}{=} \log(\hat{t}_{ij}/t_{ij})$ collected from the one-hour-ahead DCRNN forecasts. The parameter α scales the empirical forecast-error magnitude: $\alpha = 1$ corresponds to the prediction error level extracted from the trained DCRNN (used in the main experiments), $\alpha = 0$ gives the perfect-foresight benchmark, and $\alpha > 1$ is used only in the forecast-sensitivity stress tests.

5. Numerical Results

The following results concern a *single* decision slot; accordingly, the experiments assess the single-slot redesign rather than the multi-period rolling trigger of Rule 1.

5.1 Experimental setup and metrics

Mandl's Swiss network, initially used by Mandl (1980), consisting of 15 nodes, 21 links, and a symmetric demand of 15,570 passenger trips, is employed as a canonical benchmark for comparison with existing PT network design methods (Ahern et al., 2022). To further assess the method's applicability under realistic operating conditions, we consider a real-world urban public transport network in a region in Beijing, China, recently analyzed by Wu et al. (2024), which comprises 301 bus stops, 3,758 road sections, and 29 bus lines. For Mandl's network, the maximum fleet size ($V_{\max}$) is capped at 40, following the service-frequency setting of Ahern et al. (2022) to maintain comparable headways. For the Beijing network, $V_{\max}$ is set to 150 to accommodate a constant 12-minute headway across this large PT system.

For *Mandl's Swiss network*, the current network design approaches are represented by the following comparison networks:

- A static integrated line-frequency design $\mathcal{G}_{\text{PT}}^{\text{ahe}}$, following Ahern et al. (2022), which jointly optimizes passenger travel time and operational cost.
- A frequency-only adjustment baseline, denoted by $\mathcal{G}_{\text{PT}}^{\text{freq}}$, in which line structures are fixed and only service frequencies are adjusted.

To select representative compromise solutions under fluctuating travel time, we compute the solutions on the Pareto front via the proposed algorithm. Then, we compute the Euclidean distance from each of those solutions to an ideal point representing the median observed values in the Pareto front, across the three objectives. If not otherwise specified, the considered network design $\mathcal{G}_{\text{PT}}^{\text{adpt}}$ that we analyze in the following is the one with the objectives closest to that ideal point.

To evaluate the adaptive network under uncertain travel times, we report three application-specific performance indicators aligned with the objectives in (4):

- **20% improv. share (%) ($\mathbf{IS}_{20}$):** The percentage of OD pairs that achieve a travel time reduction of at least 20% compared to the original PT network.

- **Operational Cost Reduction (OCR):** percentage reduction in total vehicle-hours operated relative to the original PT network (the operational-cost proxy defined in §2.3.2).
- **Line Overlap (LO):** Jaccard-based percentage overlap (2) between the edge sets of the original and adaptive PT networks.

All reported metrics are computed under the realized link travel times, for both the original and the adaptive network (so t_k^0 in (6) is a realized travel time).

5.2 Main performance results on Mandl's benchmark

Table 1: Performance comparison of full network redesign and fixed-line frequency adjustment for the 4-line Mandl instance.

Method	Metric	CV=0.05	CV=0.25	CV=0.50	CV=1.00
Frequency Only	20% improv. share (%)	3.96 ± 2.94	8.09 ± 4.35	12.94 ± 6.92	16.47 ± 9.53
	Op. Cost Red. (%)	12.85 ± 4.66	14.86 ± 5.78	19.27 ± 9.04	29.40 ± 15.04
	Line Overlap (%)	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
Ours	20% improv. share (%)	8.25 ± 4.65	15.40 ± 5.66	24.63 ± 7.91	30.57 ± 11.60
	Op. Cost Red. (%)	13.28 ± 7.37	16.86 ± 6.74	24.06 ± 12.01	39.20 ± 12.94
	Line Overlap (%)	83.79 ± 7.73	78.83 ± 6.65	77.53 ± 5.27	77.26 ± 5.19

The 4-line static line-frequency solution is taken from Ahern et al. (2022); entries report mean $\pm$ standard deviation over 30 realizations.

Table 1 (4-line instance; other line counts are reported in the extended report, Guo et al., 2026) summarizes average performance across 30 independent realizations. As travel-time variability increases, both 20% Improvement Share (IS_{20}) and Operational Cost Reduction (OCR) rise systematically, reflecting the increasing value of adaptive redesign under highly fluctuating conditions. Across all variability levels and redesign schemes, Line Overlap (LO) remains high, indicating that these improvements result from targeted adjustments rather than wholesale restructuring.

The frequency-only baseline $\mathcal{G}_{\mathrm{PT}}^{\mathrm{freq}}$ improves service under variability, but it captures substantially less of the OD-level improvement than full redesign; for example, at $cv = 1$, its 20%-improvement share is roughly half that of our method.

We now test whether the proposed proactive redesign is driven by the intrinsic variability of link travel times rather than by forecast errors. To this aim, we hold all realized travel times $\{t_{ij}^{\omega}\}$ fixed and vary only the error-scale parameter α in Eq. (5). When variability is high, the Pareto front issued by the proposed optimization changes little when varying the forecast error factor α up to moderate amplification ($\alpha = 1.5$) and degrades only under severe amplification (see the extended report, Guo et al., 2026). The method's gains are therefore driven mainly by the redesign opportunities that variability creates, not by a dependence on high-precision forecasts.

5.3 Large-scale validation on the Beijing network

We compare our method against a *stop-skipping* operational baseline on the Beijing network. For each variability level, links whose predicted travel time exceeds the original by more than a preset threshold are classified as degraded. The stop-skipping strategy then removes intermediate stops adjacent to degraded links whenever a direct shortcut is available, while leaving the overall line structure and vehicle allocation unchanged.

In addition to the previous metrics, we also compute the following.

- **Avg. Travel Time Improvement (Avg.TTI):** percentage reduction in aggregate travel time across

Table 2: Comparison of our method and stop-skipping under varying travel-time variability on the Beijing network.

Method	Thr. Skip threshold	CV Coeff. var.	LO Line Overlap (%)	OCR Op. cost red. (%)	Avg.TTI Travel-time impr. (%)	Unreach. OD Unreachable OD (%)	Unserv. Dem. Unserved demand (%)
Stop-skip.	25%	0.50	47.58	6.95	14.08	35.28	32.99
		1.00	42.18	12.71	14.71	38.93	36.04
	50%	0.50	56.13	8.69	19.42	28.63	27.56
		1.00	47.83	12.39	19.71	35.28	32.99
	100%	0.50	78.57	6.68	17.77	12.23	11.61
		1.00	55.88	13.49	15.11	28.05	26.84
Ours	–	0.00	90.08	0.50	11.40	–	–
		0.05	87.34	1.53	11.60	–	–
		0.25	86.51	3.85	16.99	–	–
		0.50	86.39	12.02	18.44	–	–
		1.00	85.42	22.56	25.78	–	–

CV: coefficient of variation. Thr.: skip threshold used by the stop-skipping baseline. LO: Line Overlap. OCR: Operational Cost Reduction, measured as the reduction in total vehicle-hours; see §2.3.2. TTI: Travel-Time Improvement; see (6). Unreach. OD and Unserv. Dem. report the fraction of OD pairs and demand, respectively, whose connectivity is broken by stop-skipping. These metrics are not applicable to our method, which removes no stops and preserves full OD connectivity. Stop-skipping rows for $cv \in \{0, 0.05, 0.25\}$ are reported in the extended report (Guo et al., 2026).

all OD pairs k relative to the original network:

$$\text{Avg.TTI} = \frac{\sum_{k\in\mathcal{K}}(t_k^0 - t_k^*)}{\sum_{k\in\mathcal{K}} t_k^0} \times 100\%, \text{ where } t_k^0, t_k^* \text{ are original and adaptive travel times.} \quad (6)$$

The results in Table 2 show that the original network exhibits some inefficiencies: the solution of our optimization yields an 11.40% Avg. TTI even at $cv = 0$. This gain at $cv = 0$ does not stem from prediction-driven adaptation: with no travel-time variability, there is nothing to forecast or to adapt to. The genuine value of *adaptation* is therefore the additional gain observed as cv increases, beyond this one-time static correction. The benefit grows with variability: at higher cv, the algorithm produces designs increasingly different from the static original, achieving both travel-time improvements and line shortening, which in turn lowers operational cost.

Table 2 exposes a basic tension in stop-skipping: operational benefit comes at the cost of connectivity. Aggressive thresholds improve travel time on the reachable network but disconnect a large share of OD pairs as variability grows, whereas conservative thresholds keep connectivity yet act too rarely to help and still lose connectivity at high variability.

Our method achieves larger travel-time gains for two structural reasons. First, it redesigns entire line topologies rather than dropping individual stops, so vehicles can bypass congested corridors and serve demand more directly; this reaches reductions that stop-level changes cannot, since the bottleneck is usually the line's overall routing rather than a single link. Second, it jointly reallocates the fleet across lines, redeploying vehicles freed from shortened routes where they help most, while stop-skipping keeps the fleet fixed.

6. Conclusion

This paper systematically studied the potential of proactive bus line redesign, targeted at limited parts of the network, in maintaining user and operator performance in the case of severe traffic fluctuations. Results indicate that, faced with non-negligible link travel-time variability, the proposed approach yields OD-level trip travel-time reductions, while achieving operational cost savings over the selected baselines and scenarios considered, and preserving a high degree of structural similarity with the baseline network. Overall, the gains grow with travel-time variability and come with high line overlap, indicating that the benefit does not rely on large structural change.

A natural extension is to enrich the forecast-error model with explicit spatial and temporal correlation, allowing network-wide common shocks to be represented.

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